



FINAL PROJECT

PREDICTION OF SODIUM HYPOCHLORITE AS COOLING TOWER NEEDS

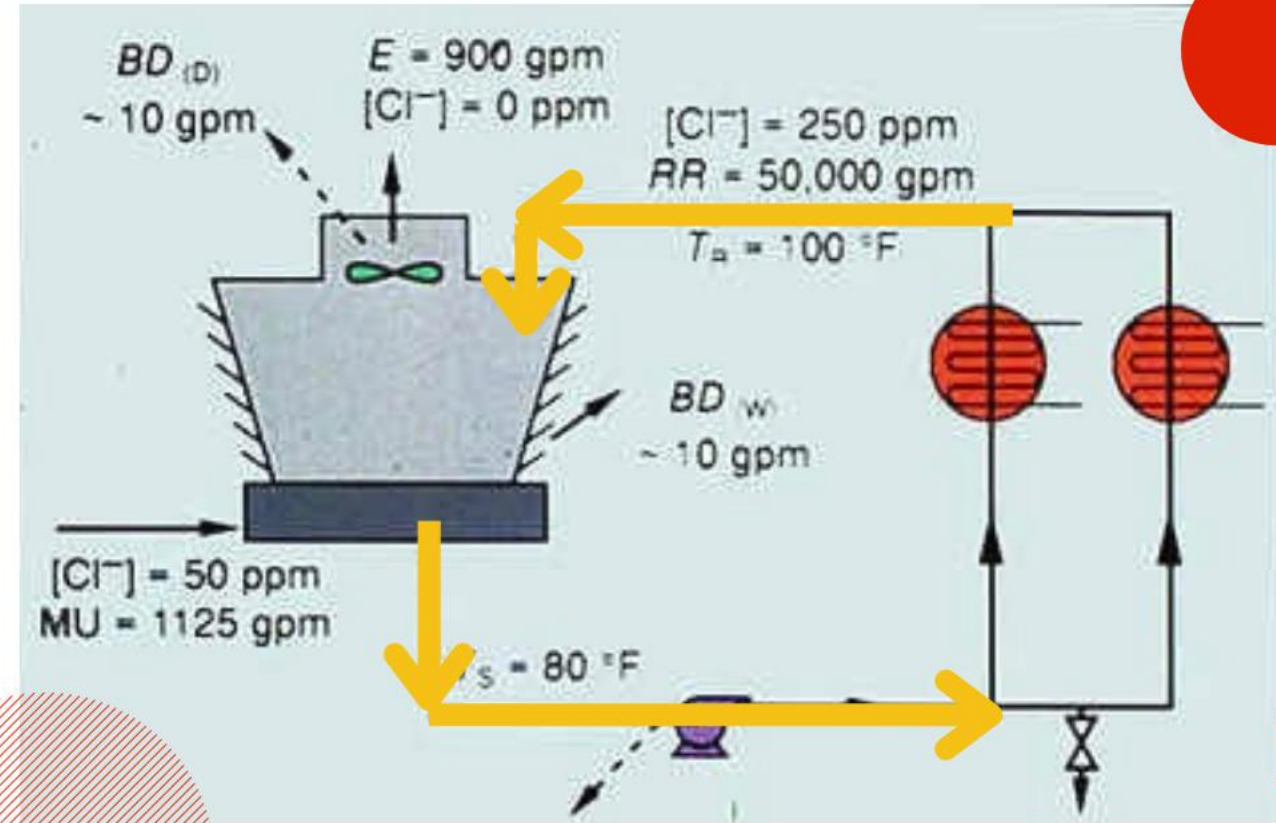
Andrew NG

Background & Problem Statement



A device that removes waste heat from a coolant stream, usually water, by cooling it to a lower temperature.

Cooling towers can use evaporation to cool the water, or they can use air to cool the water.



Background & Problem Statement



Cooling Tower

Cooling tower need maintenance to prevent problems and maintain good heat transfer in a process. Common problems in cooling tower mentioned below:

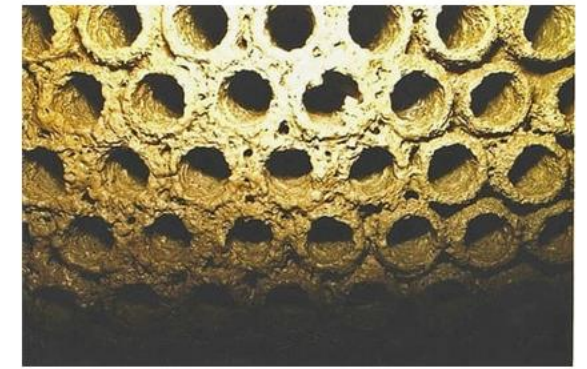


Corrosion

Biofilm

Scaling

Micro organism growth is the one to be particularly concerned about as it can lead to corrosion and biofilm.



Micro organism



Oxidizing Biocide

Dose automatically



Non-oxidizing
Biocide

Dose manually

Controlled by:

ORP (Oxidation
Reduction Potential)
which **indirectly**
represent FRC (Free
Residual Chlorine)

Affect:

1. Chloride concentration in system
2. Cooling tower conductivity
3. Water consumption
4. Total amount bacteria



Background & Problem Statement

Oxidizing Biocide

Most common oxidizing:



Sodium Hypochlorite
(NaOCl)

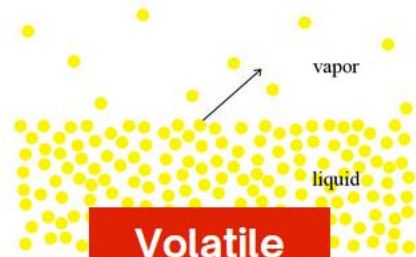


OCl(-) is an active content that aims to damage the structure and function of microorganisms through the oxidation process.



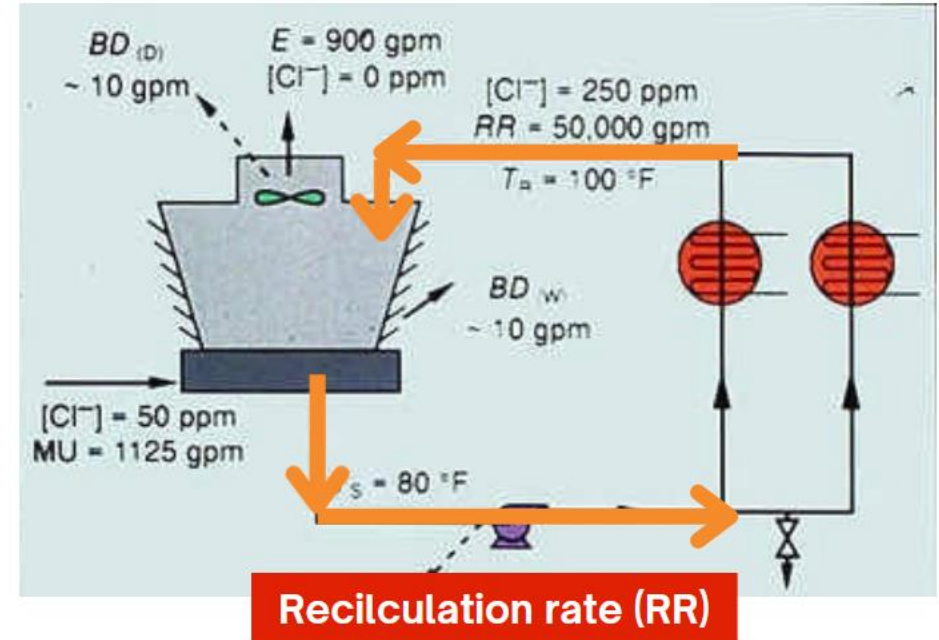
Masive amount usage

(Depends on recirculation rate of cooling tower)



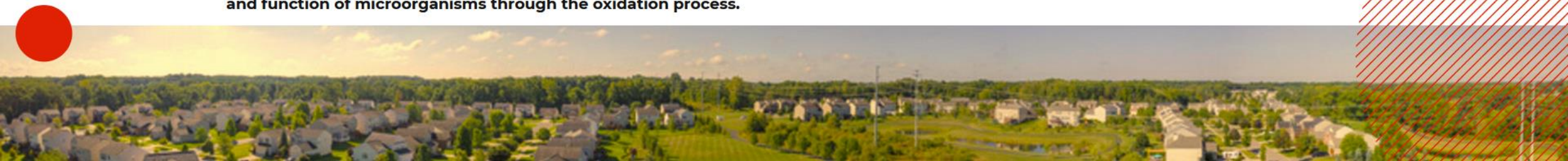
Volatile

*Longer storage times = less concentration of NaOCl
*after opened



Recirculation rate (RR)

High RR of cooling tower = High NaOCl usage



Background & Problem Statement

Problem Statement



sodium hypochlorite

**Sodium Hypochlorite
(NaOCl)**



Too much

Overstock

Volatile

Redundant

Too less



**High bacteria
growth**

**Potential
corrosion &
bio film
increase**

Goals



sodium hypochlorite

**Predict specific amount of Sodium
Hypochlorite for cooling tower treatment**



Data Collection & Preparation

● Feature

```
Index(['ID CT', 'FUNCTION', 'MATERIAL', 'DAY_RUN', 'RR', 'NAOCL',  
      'AVG_T_RETURN', 'CONDUCT LIMIT', 'AVG_FRC', 'AVG_CHLORIDE',  
      'BACTERIA_STATUS'],  
      dtype='object')
```

● Numeric Data

```
['DAY_RUN', 'RR', 'AVG_T_RETURN', 'AVG_FRC', 'AVG_CHLORIDE']
```

● Output Data

```
['DAY_RUN',  
 'RR',  
 'AVG_T_RETURN',  
 'AVG_FRC',  
 'AVG_CHLORIDE',  
 'CONDUCT LIMIT',  
 'BACTERIA_STATUS',  
 'FUNCTION',  
 'Cuprum',  
 'Galvanized',  
 'Stainless steel']
```

```
BACTERIA_STATUS  
2          177  
1           42  
Name: count, dtype: int64
```

● Conduct Limit

```
CONDUCT LIMIT  
3           80  
2           79  
1           60  
Name: count, dtype: int64
```

● Shape

```
(175, 12)
```

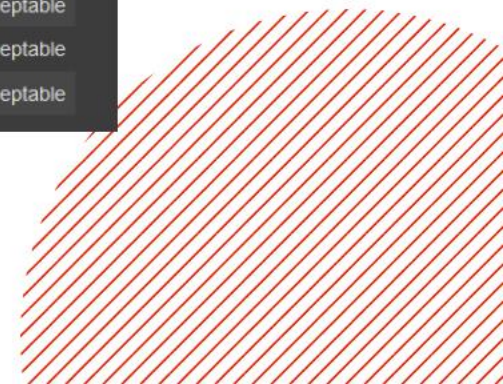


EDA & Preprocessing

Import Library

```
[ ] import pandas as pd
import numpy as np
import ast
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import OneHotEncoder, StandardScaler, RobustScaler, MultiLabelBinarizer
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression, Lasso, Ridge
from sklearn.ensemble import StackingRegressor
from xgboost import XGBRegressor
from lightgbm import LGBMRegressor
from sklearn.svm import SVR
from sklearn.neighbors import KNeighborsRegressor
from sklearn.tree import DecisionTreeRegressor
from sklearn.ensemble import RandomForestRegressor
from sklearn.model_selection import GridSearchCV
from sklearn.metrics import mean_absolute_error, mean_absolute_percentage_error, r2_score, median_absolute_error, mean_squared_error
from sklearn.base import BaseEstimator, TransformerMixin
```

	ID CT	FUNCTION	MATERIAL	DAY_RUN	RR	NAOCL	AVG_T_RETURN	CONDUCT	LIMIT	AVG_FRC	AVG_CHLORIDE	TOTAL BACTERIA	BACTERIA_STATUS
0	Cooling Tower Ref A	PROCESS	Stainless steel	30	400	297.69	32.1	1200	0.21	158.07		1000	Acceptable
1	Cooling Tower Fract B	FRACTIONATION	Stainless steel, Cuprum, Galvanized	30	2400	1997.31	35.4	1500	0.19	293.87		1000	Acceptable
2	Cooling Tower Ref C	PROCESS	Stainless steel	30	450	716.54	39.3	1200	0.19	185.07		1000	Acceptable
3	Cooling Tower Fract D	FRACTIONATION	Stainless steel, Cuprum, Galvanized	30	700	1076.54	31.8	700	0.26	127.24		1000	Acceptable
4	Cooling Tower Refra E	FRACTIONATION	Stainless steel, Cuprum, Galvanized	30	2496	1041.92	32.6	1500	0.14	240.24		10000	Acceptable



EDA & Preprocessing

Keterangan Feature:

Detail feature:

- ID CT = Identitas Cooling Tower
- FUNCTION = Tipe proses yang didinginkan oleh cooling tower
- MATERIAL = Material pipa yang dilewati oleh cooling water
- DAY_RUN = Hari digunakannya cooling tower
- NAOCL = Jumlah NAOCL dalam kg yang dibutuhkan dalam 1 bulan
- CONDUCT_LIMIT = Limit conductivity terhadap design cooling tower
- AVG_FRC = rerata konsentrasi Free Residual Chlorine (ppm)
- AVG_CHLORIDE = rerata konsentrasi chloride (ppm)
- TOTAL_BACTERIA = Total bakteri dalam cooling tower
- RR = Recirculating Rate Cooling Tower (m3/h)
- BACTERIA_STATUS = Kondisi ditoleransi/tidaknya jumlah bakteri yang ada pada cooling tower

Informasi tambahan:

- AVG_FRC ideal adalah 0.2-0.5

Beberapa feature yang bisa dikombinasi:

- Material X AVG_Chloride
- AVG_FRC X Total bacteria
- AVG_FRC X Bacteria status
- RR X AVG_FRC = untuk melihat apakah semakin besar RR menunjukkan kecenderungan semakin sulit menjaga nilai VG_FRC

```
[ ] df.describe()
```

	DAY_RUN	RR	NAOCL	AVG_T_RETURN	CONDUCT_LIMIT	AVG_FRC	AVG_CHLORIDE	TOTAL_BACTERIA
count	220.000000	220.000000	220.000000	220.000000	220.000000	220.000000	220.000000	220.000000
mean	27.900000	1319.272727	796.932109	36.054545	1172.727273	0.320136	167.074068	28404.545455
std	5.391436	849.473121	496.167222	4.090163	317.210758	0.212142	123.924778	75349.080933
min	0.000000	350.000000	0.000000	31.800000	700.000000	0.000000	39.750000	0.000000
25%	28.000000	450.000000	433.920000	32.100000	700.000000	0.190000	120.670000	1000.000000
50%	30.000000	1092.000000	711.710000	35.400000	1200.000000	0.260000	155.720000	10000.000000
75%	31.000000	2400.000000	1050.045000	39.300000	1500.000000	0.380000	187.317500	10000.000000
max	31.000000	2500.000000	3080.800000	45.400000	1500.000000	1.310000	1800.360000	1000000.000000

```
[ ] df.isnull().sum()
```

```
ID CT      0
FUNCTION    0
MATERIAL     0
DAY_RUN     0
RR          0
NAOCL       0
AVG_T_RETURN 0
CONDUCT_LIMIT 0
AVG_FRC     0
AVG_CHLORIDE 0
TOTAL_BACTERIA 0
BACTERIA_STATUS 0
dtype: int64
```

```
[ ] df['MATERIAL'].value_counts()
```

```
MATERIAL
Stainless steel      99
Stainless steel, Cuprum, Galvanized  80
Stainless steel, Galvanized  40
Name: count, dtype: int64
```

```
[ ] df['DAY_RUN'].value_counts()
```

```
DAY_RUN
30    77
31    71
29    14
28    12
22     8
24     6
18     5
20     4
8       4
25     4
27     3
26     2
12     2
15     1
14     1
19     1
16     1
13     1
6       1
23     1
Name: count, dtype: int64
```

```
[ ] df['TOTAL_BACTERIA'].value_counts()
```

```
TOTAL_BACTERIA
10000    108
1000     69
100000   41
1000000    1
Name: count, dtype: int64
```

```
[ ] df['BACTERIA_STATUS'].value_counts()
```

```
BACTERIA_STATUS
Acceptable    177
Not Acceptable  42
Name: count, dtype: int64
```

```
[ ] df['ID_CT'].value_counts()
```

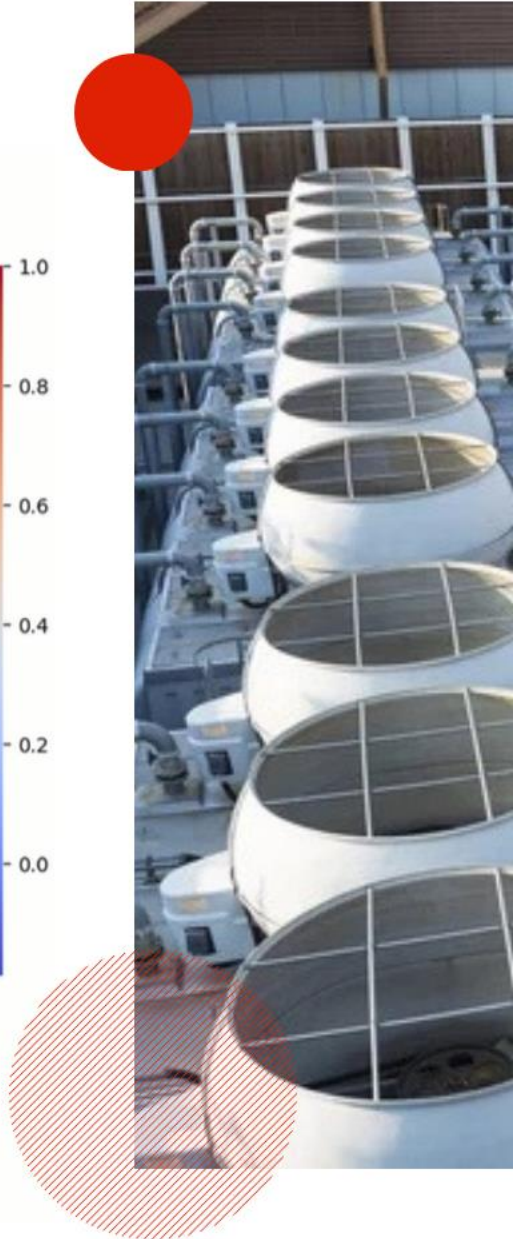
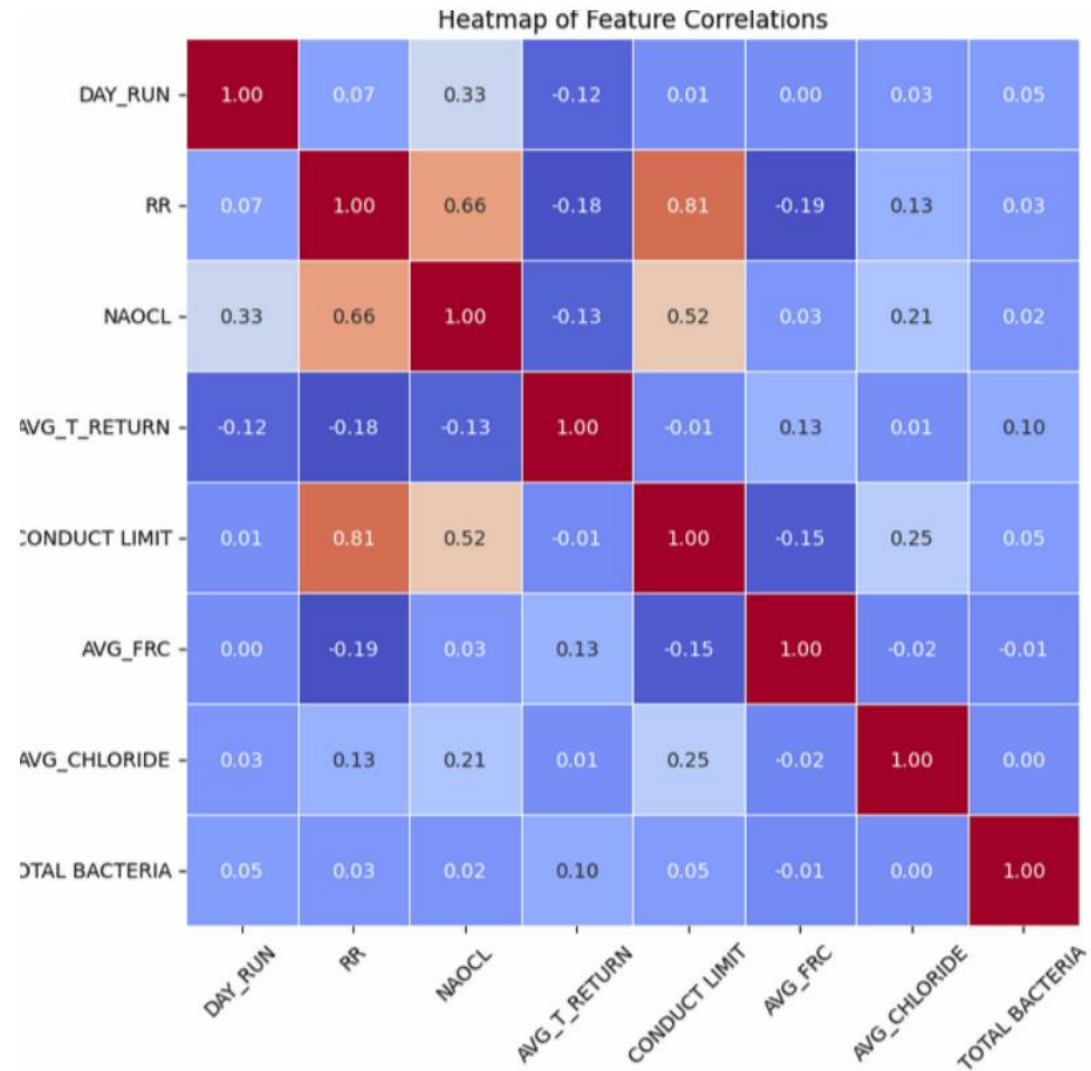
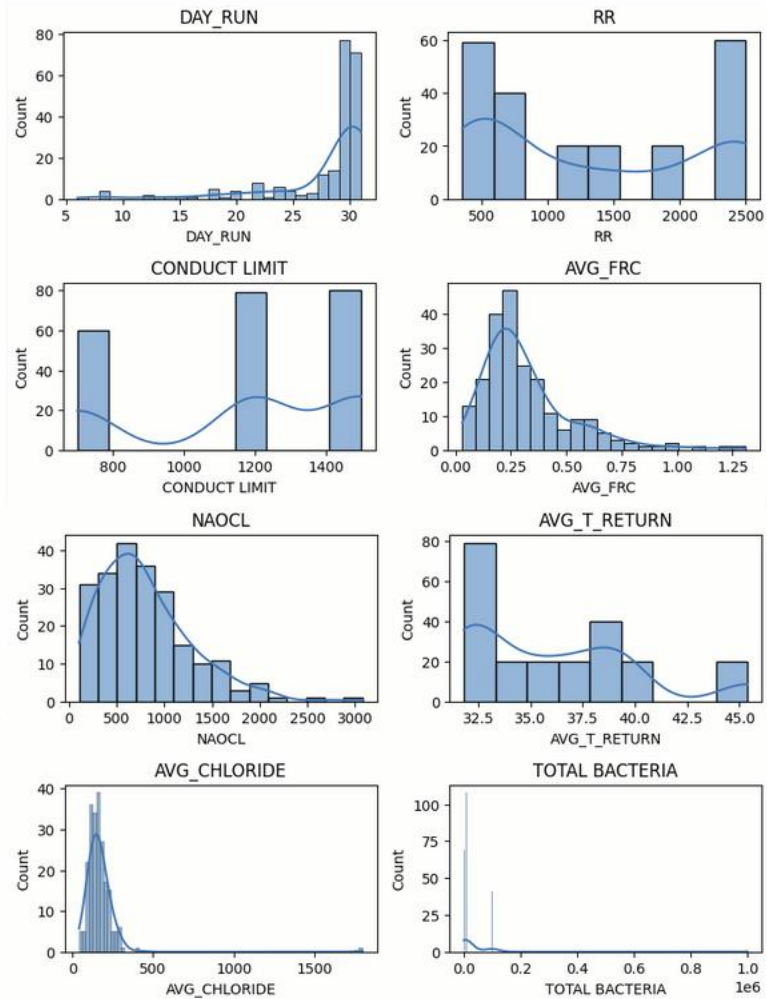
```
ID_CT
Cooling Tower Fract B    20
Cooling Tower Ref C     20
Cooling Tower Fract D    20
Cooling Tower Ref H     20
Cooling Tower Refra E    20
Cooling Tower Ref F     20
Cooling Tower Fract G    20
Cooling Tower Fract J    20
Cooling Tower Fract I    20
Cooling Tower Ref K     20
Cooling Tower Ref A     19
Name: count, dtype: int64
```

```
[ ] df['CONDUCT_LIMIT'].value_counts()
```

```
CONDUCT_LIMIT
1500    80
1200    79
700     60
Name: count, dtype: int64
```

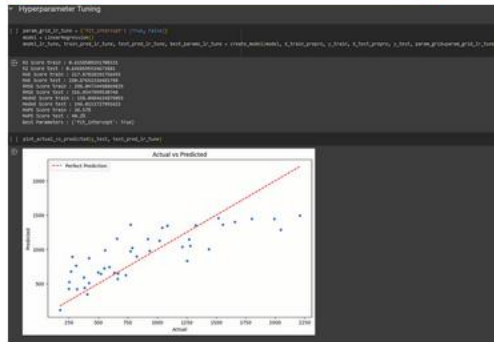


EDA & Preprocessing

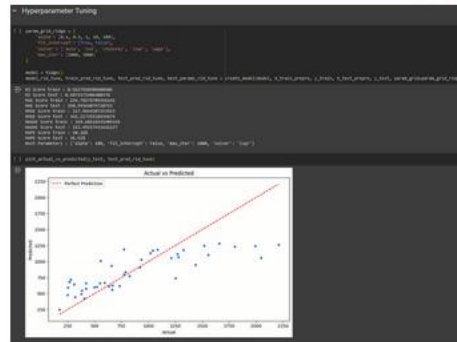


Modeling

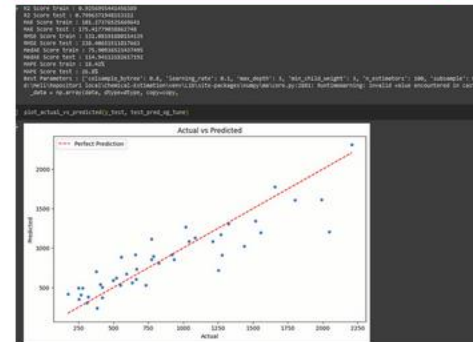
Linear Regression



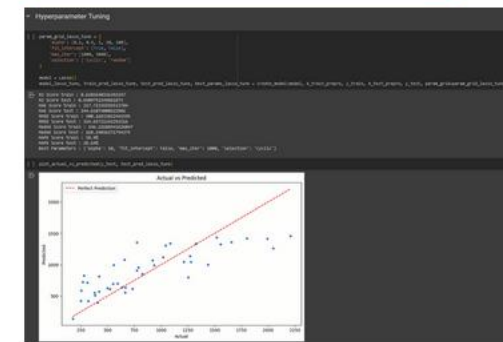
Ridge



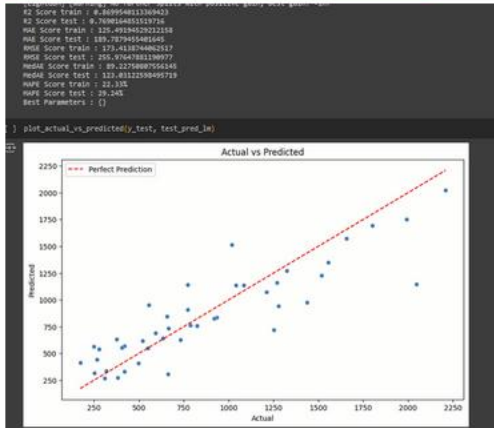
XG Boost



Lasso



LGBM

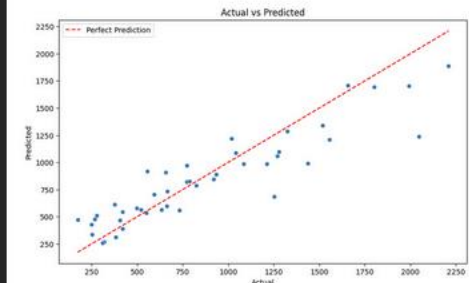


Random Forest



EnSamble Stucking

```
R2 Score train : 0.891112998666991
R2 Score test : 0.81336663163732
MAE Score train : 114.80114466439346
MAE Score test : 168.64638501998795
RMSE Score train : 158.6804341006582
RMSE Score test : 230.09342830537796
MedAE Score train : 86.05861301781658
MedAE Score test : 110.79368687418292
MAPE Score train : 20.08%
MAPE Score test : 25.83%
Best Parameters : {}
```



Deployment

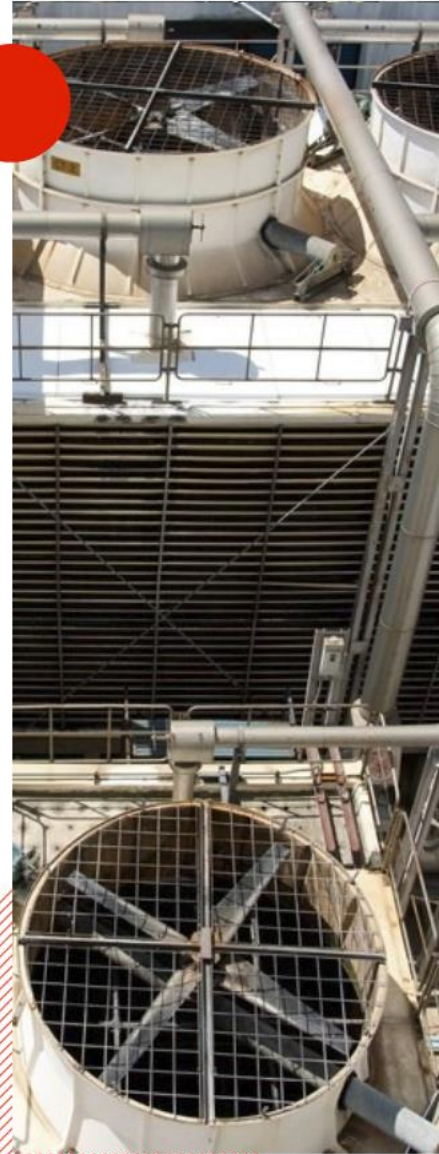
```
1 from flask import Flask, request, render_template
2 import numpy as np
3 import pandas as pd
4 import sys
5 import os
6
7 # Tambahkan root folder Spam-Message-Detection ke sys.path
8 sys.path.insert(0, os.path.dirname(os.path.dirname(os.path.abspath(__file__))))
9
10 from src.predict_pipeline import PredictionPipeline, CustomData
11
12 application = Flask(__name__)
13 app = application
14
15 # Route for first page
16 @app.route("/")
17 def index():
18     return render_template('index.html')
19
20 # Route Predict page
21 @app.route("/predict", methods=['GET', 'POST'])
22 def predict_datapoint():
23     if request.method == 'GET':
24         return render_template('predict_page.html')
25     else:
26         data = CustomData(
27             DAY_RUN=float(request.form.get('DAY_RUN')),
28             RR=float(request.form.get('RR')),
29             AVG_T_RETURN=float(request.form.get('AVG_T_RETURN')),
30             AVG_FRC=float(request.form.get('AVG_FRC')),
31             AVG_CHLORIDE=float(request.form.get('AVG_CHLORIDE')),
32             CONDUCT_LIMIT=int(request.form.get('CONDUCT_LIMIT')),
33             BACTERIA_STATUS=request.form.get('BACTERIA_STATUS'),
34             FUNCTION=request.form.get('FUNCTION'),
35             MATERIAL=request.form.getlist('MATERIAL') # Ambil sebagai list
36         )
37
```

```
df_pred = data.get_data_input()
print(df_pred)
print(f'Before prediction')

predict_pipeline = PredictionPipeline()
results = predict_pipeline.prediction(df_pred)
results = np.round(results, 2)
print(f'After prediction')

return render_template('predict_page.html', results=results[0])

if __name__ == '__main__':
    app.run(debug=True, port=os.getenv("PORT", default=5000))
```



Deployment

Prediction Form

DAY_RUN

RR

AVG_T_RETURN

AVG_FRC

AVG_CHLORIDE

CONDUCT_LIMIT

BACTERIA_STATUS

Acceptable

FUNCTION

FRACTIONATION

MATERIAL

Stainless steel
Cuprum
Galvanized

Predict

web link:

<https://chemical-trend-estimate-production.up.railway.app>

